

**SMART INDIA
HACKATHON
2026**

The logo is a central graphic on a light gray background. It features a stylized brain shape divided vertically. The left half is orange and filled with a circuit board pattern. The right half is green and filled with white binary code (0s and 1s). Above the brain, the letters 'CHI' are written in a bold, black, serif font. Below the brain, the letters 'SIH' are written in a bold, gray, sans-serif font, with a horizontal bar above them. The entire logo is framed by a large, light gray hexagon.

Team Name: Team Vincera

SMRITI SATHI

Proposed Solution: games and daily memory assistance

23 games. Orchid Pairs matches hidden cards. Memory Tray asks which item disappeared after a 10-second view. [1]

Familiar daily life. Order routines in Daily Life Sequence. Recognise regional foods. Share memories through Festival Tales. [2, 3]

Problem fit. Six language choices, large controls, offline activities, medicine/routine reminders and a local caregiver view.

Innovation. Regional content and gentle cues combine with difficulty that responds to answers.

Technologies. React, TypeScript, Vite and Capacitor. IndexedDB stores local data. A service worker caches core assets.

Current game loop. Choose local content, record an answer, update domain ability, then set the next round's difficulty.

AI personalisation. Local LinUCB recommends the next game. The separate 25-feature neural question engine still needs gameplay integration.

Optional online services. A server proxy connects chat and speech providers. Core play continues offline.

Feasibility. Games, local adaptation, reminders and caregiver views run on standard mobile/tablet hardware.

Challenges and risks. Speech support varies by language. Notifications need permissions. Clearing app data can erase local records.

Strategies. Use cached audio and text fallbacks. Check permissions. Add encrypted backup, authentication and sync before wider deployment.

Next validation. Native-speaker review, accessibility sessions with older adults, then a clinician-supervised pilot.

For older adults. Familiar activities aim to encourage daily engagement and shared play with family.

For caregivers. One local view shows completed sessions, reminder logs and activity trends.

For remote households. Core use needs no continuous connection or special hardware. Optional cloud services add running costs.

How we will measure benefit. Measure completion, reminder acknowledgement, caregiver effort and usability in a pilot. Clinical benefit remains unproven.

[1] Chan et al. (2024), npj Digital Medicine
Memory training. Design link: Pairs and Tray. Dementia evidence
is limited.

[2] Woods et al. (2023), Cochrane
Cognitive stimulation. Design link: varied activities. Evidence is
mostly from groups.

[3] Elfrink et al. (2021), PLOS ONE
Life Story Book trial. Design link: Festival Tales. Most outcomes
were not significant.

Scope: 2021–2026 research. These papers do not validate Smriti
Sathi